

Stephen O'Neil

Full-Stack Web Developer

11+ years frontend | 8+ years backend & DevOps
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A senior developer & technical advisor, modernizing Canada's public health data platforms & infrastructure.

Previously lead developer on GC InfoBase, a React SPA client backed by a serverless GraphQL API.
(canada.ca/gcinfobase, github.com/TBS-EACPD/infobase)

Education

Honours B.Sc. Physics-Mathematics, uOttawa
2017, Cum Laude

Certifications

Google Cloud Certified - Professional Cloud Architect
(tinyurl.com/oneil-gc-cert)

Languages

- TypeScript & JavaScript
- HTML ➤ CSS & Sass
- Python ➤ SQL
- sh / Bash / POSIX shells

Libraries & Frameworks

- React ➤ Express.js
- GraphQL ➤ Apollo
- Yoga ➤ Lodash
- React Testing Library
- Jest ➤ Puppeteer
- Cypress ➤ Axe-core
- Astro ➤ NextJS
- Webpack ➤ Vite
- Django ➤ pytest
- OpenTelemetry

Resume

Senior Developer & Technical Advisor (01/2023 - Present)

Digital Transformation Branch, Health Canada ————— 07/2024 - Present

- Architected and lead development on the [Interoperable Immunization Data Initiative](#) proof of concept, to demonstrate both simulated cross-jurisdictional data exchange between HL7 FHIR APIs and practical federated data mesh concepts
 - Implemented as four types of microservice, with instances deployed in a mesh pattern, and three dashboards for demo purposes. Microservices primarily written in TypeScript as lightweight Express.js servers
 - Employed a state machine pattern built on Redis job queues to provide fault tolerance, retry logic, and rollback capabilities for critical transaction services
- Iterated on a proof of concept for [safer data collection methods](#). Enabled existing Excel based exchanges through the platform, with the added benefit of schema validation and value extraction occurring in the sender's browser, to shift the security boundary left and to obviate manual, async, email exchange based data QA
 - Built an Express.js backend providing authn & authz over top of a GraphQL (Yoga) API for user creation and versioning of custom schemas, and for the collection of data against these custom schemas. Also built the corresponding React frontend

Data, Surveillance and Foresight Branch, Public Health Agency of Canada ————— 06/2023 - 07/2024

- Contributed logging utilities to the [branch's internal Django library](#), enabled standardized, enriched, and security policy compliant logs in both Google Cloud & Azure environments
- Drafted infrastructure and security architecture for the branch's first Kubernetes-based Django application deployment, lead implementation work
 - Coordinated with project managers, application developers, and IT security to collect and track business and technical requirements
 - Achieved a production-ready deployment, obtained CIO Authority to Operate, and provided lessons-learned for future kubernetes use within the organization

Science and Parliamentary Infrastructure Branch, Public Services and Procurement Canada ————— 01/2023 - 06/2023

- Advised on technical aspects of a branch-wide data policy refresh
- Triaged a portfolio of ~20 projects left in states varying from prototype to pilot
 - Identified and decommissioned 10+ dead projects, directly reduced cloud bills, maintenance burden, and potential risks & attack surfaces
 - Matured the remaining applications and infrastructure; eliminated manual deploy steps, implemented automated rollback capabilities & hardened configurations

Tools & Software

- Git
- Linux
- VSCode
- MongoDB
- Kubernetes
- NodeJS
- Docker
- PostgreSQL
- Redis

Platforms

- Google Cloud Platform
- DigitalOcean
- GitHub Actions
- CircleCI

Other Skills & Competencies

- Architecture
- Application security
- Accessibility
- Strategic leadership
- Automation
- Mentorship

Resume (Cont.)

From Developer to Lead Developer (09/2017 - 01/2023)

Expenditure Management Sector, Treasury Board of Canada Secretariat

- Lead the [GC InfoBase](#) project beginning in Summer 2018; held responsibility for the project's technical direction, overall design, and for the team's developer talent
 - Developed and maintained the project's large React frontend
 - E.g. through gradual refactors to legacy code, brought the average turn around of routine site updates [from a baseline of 5 business days to an average of 0.13](#)
 - Built backend microservices to enable new features in the core application
 - E.g. built a versatile template based form service, used to collect, validate, store, and alert the team across multiple lines of in-app user feedback
 - Architected and administered infrastructure and operations for the project
 - E.g. transitioned hosting from tenancy in an external team's servers to cloud platforms & an in-house DevOps culture. Brought the deployment process from days to minutes while providing a capable backend + flexible & faster hosting
 - Developed tooling to multiply team productivity and reinforce best practices
 - E.g. implemented automated end-to-end WCAG 2.1 accessibility testing for CI
 - Championed code standards, review practices, testing, CI/CD, and working in the open
 - E.g. sold stakeholders on open sourcing the project; was awarded in recognition of the direct positive impacts one year later
 - Hired, supervised, and mentored 3 junior developers and 10+ CO-OP students
- Built the data collection tool used for the centralized tracking of COVID-19 related expenditures across all government departments, on an accelerated time-line
 - Rapidly picked up Python, Django, an existing internal platform's code & tooling, and the initial business requirements; began implementation by day 6
 - Individually developed the new collection tool off of the existing platform, with robust user and admin workflows, and still met the ambitious launch date of 3 months from project initialization
 - Contributed lasting improvements to the existing platform's testing practices

Student & Part-time Developer (01/2015 - 09/2017)

Office of the Chief Human Resources Officer, Treasury Board of Canada Secretariat

- Self-taught JS, HTML, CSS, and Git; developed the early HR data portions of GC InfoBase